An Efficiency-Aware Convolutional Neural Network for Medical Image Classification

Group 11
Hung Siu Wang 57148331
Chan Tsz Hin 58539653
Matteo Zabban 40172615
Choi San Ho 57849436

April 25, 2026

Abstract

This project investigates the use of convolutional neural networks for multi-class skin lesion classification on the ISIC 2019 dataset, with a specific focus on efficiency-aware model design. Skin lesion analysis is a clinically relevant task, as early and accurate detection of suspicious lesions can support dermatological diagnosis and improve patient outcomes. However, the problem is challenging due to strong class imbalance, the presence of uncertain labels such as the `UNK` category, substantial variability in image quality, and the risk of data leakage caused by multiple images of the same lesion.

To address these issues, the project first performs a detailed analysis of the dataset structure and its main limitations. The `UNK` class is excluded to define a cleaner and more coherent classification task, while lesion-aware splitting is adopted to avoid overly optimistic validation results. In particular, a `StratifiedGroupKFold` strategy is used to preserve class balance as much as possible while ensuring that images of the same lesion do not appear in both training and validation sets. In addition, preprocessing and normalization steps are introduced to reduce the impact of heterogeneous image quality and acquisition conditions.

On this basis, the study develops and evaluates a convolutional neural network architecture designed to balance predictive performance and computational efficiency. Rather than focusing only on raw accuracy, the project emphasizes balanced evaluation metrics such as macro F1-score, which are more appropriate in the presence of severe class imbalance. Experimental results are analyzed both quantitatively and qualitatively, with attention to class-specific behavior and model robustness.

Overall, the project shows that skin lesion classification on ISIC 2019 is not only a challenging computer vision problem but also a task where methodological choices strongly affect the reliability of the results. The final outcome is an efficiency-aware CNN pipeline that provides a realistic and careful approach to dermatological image classification under practical dataset constraints.

Contents

1 Introduction **2**

2 The Dataset **4**

3 Major Issues in the Dataset **7**
3.1 UNK Class and Class Imbalance 7
3.2 Image Quality Variability 8
3.3 Lesion ID and Data Leakage Risk 10

4 Baseline Model Comparison **12**
4.1 ResNet-18 Baseline 12
4.2 MobileNetV2 Baseline 13
4.3 EfficientNet-B0 Baseline 14
4.4 Comparison Summary 15

5 Improvement Strategy **16**
5.1 Light Data Augmentation 16
5.2 CutMix 16
5.3 Cosine Annealing Scheduler 17
5.4 Test-Time Augmentation (TTA) 18

6 Final Results **20**
6.1 Validation Confusion Matrix (TTA) 22
6.2 Validation Prediction by Class 22
6.3 Loss and F1-Score Curves 22
6.4 Classification Report and Class-wise Performance 23
6.5 Future Work 23

7 Conclusion **24**
7.1 Key Findings 24
7.2 Challenges Addressed 24
7.3 Future Directions 25
7.4 Contribution to Dermatological AI Research 26

A Contribution Report **28**

Chapter 1

Introduction

Skin lesion classification is an important problem in medical image analysis because early detection of skin cancer can make a major difference in patient outcomes. Among skin cancers, melanoma is one of the most dangerous forms, and diagnosing it at an early stage can be life-saving. In clinical practice, dermatologists often rely on dermoscopy, which makes it possible to examine skin lesions in greater detail than with the naked eye. However, even with dermoscopic images, diagnosis is not always straightforward. Different lesion types can look very similar, while lesions from the same category may appear quite different depending on the patient, the lighting conditions, the lesion size, or the image quality.

Because of these challenges, skin lesion classification has become a relevant application for deep learning, and especially for convolutional neural networks. CNNs are well suited for image tasks because they can automatically learn useful visual patterns directly from the data, from simple features such as edges and color changes to more complex structures like texture, asymmetry, and border irregularity. In recent years, these models have shown strong results in many medical imaging tasks, including dermatology. In this context, an automatic classification system is not meant to replace clinicians, but it can still be useful as a support tool for screening, prioritization, and decision assistance.

In this project, we use the ISIC 2019 dataset, which is one of the most widely used benchmarks for skin lesion analysis. There are several reasons behind this choice. First, the dataset is relatively large compared with many other medical imaging datasets, which makes it suitable for training deep learning models. Second, it includes multiple lesion categories instead of a simple benign-versus-malignant distinction, making the problem more realistic and more challenging. Third, the dataset is strongly imbalanced, which reflects the kind of difficulty that is often present in real medical data. Finally, because ISIC 2019 is widely used in the literature, it provides a meaningful reference point for evaluating our work.

At the same time, ISIC 2019 is far from being an easy dataset. Lesions belonging to the same class can vary a lot in appearance, while lesions from different classes may share similar visual characteristics. In addition, many images include artifacts such as hair, shadows, rulers, or variations in illumination and sharpness. These aspects make the classification task more realistic, but they also make it harder. For this reason, building a good model is not only a matter of choosing an architecture, but also of carefully handling preprocessing, augmentation, and evaluation.

The main goal of this project is to design and evaluate an efficiency-aware convolutional neural network for multi-class skin lesion classification. In many deep learning

studies, the focus is mainly on achieving the highest possible predictive performance, often by using larger and more computationally expensive models. However, in practice, efficiency also matters. A model that is slightly less accurate but much lighter, faster, and easier to train may be more useful in real applications, especially in settings with limited computational resources. For this reason, our objective is not simply to maximize accuracy, but to find a better balance between performance and efficiency.

This project therefore looks at skin lesion classification from a practical perspective. We are interested not only in predictive metrics such as accuracy, precision, recall, and F1-score, but also in whether the model remains simple enough to be trained and deployed in a realistic setting. In an imbalanced medical dataset, this is particularly important, because strong overall accuracy can sometimes hide weak performance on minority classes. As a result, evaluation must be interpreted carefully and should include balanced metrics as well.

Another relevant aspect of the project is interpretability. In medical applications, it is useful to understand whether a model is focusing on meaningful image regions when making predictions. For this reason, methods such as Grad-CAM can provide an additional qualitative perspective. They do not fully explain the model's reasoning, but they can still help us check whether the network is attending to plausible lesion areas rather than irrelevant background patterns or artifacts.

Overall, this project lies at the intersection of medical image classification and efficient deep learning. By working with the ISIC 2019 dataset and focusing on an efficiency-aware CNN, we aim to study whether a relatively compact model can still achieve solid results on a challenging and realistic dermatology benchmark.

Chapter 2

The Dataset

This project is based on the **ISIC 2019** skin-lesion dataset, which is designed for automated lesion classification from dermoscopic images. A major strength of this dataset is that it combines two complementary sources of information. The first component is the set of **dermoscopic images**, which represents the main input for the convolutional neural network. These images contain the visual patterns that the model must learn in order to distinguish among lesion categories, such as color distribution, border irregularity, texture, asymmetry, and local structures within the lesion. The second component is a **tabular metadata file**, which provides additional descriptive information for each sample. Even though the main focus of this project is image-based classification, the metadata remains useful for understanding the dataset composition, exploring possible biases, and performing descriptive analysis.

The image component is the core of the classification task. Each image corresponds to a single skin lesion captured under dermoscopic conditions, which generally provide better visual detail than standard clinical photography. Compared with natural-image datasets, these images are more specialized and often more difficult to interpret, since clinically meaningful differences between classes may be subtle. In addition, lesions can appear under different illumination conditions, magnifications, skin tones, and acquisition settings. Many images also contain artifacts such as hair, rulers, gel bubbles, dark corners, or uneven lighting. These factors make the task more realistic, but also more challenging for machine learning models.

Alongside the images, the dataset includes structured metadata associated with each lesion. This tabular component contains information that helps contextualize the samples and better understand the population represented in the dataset. In general, metadata fields include identifiers and patient- or lesion-level attributes such as age, sex, anatomical site, and diagnostic information. These variables are useful during exploratory data analysis because they help reveal how the dataset is distributed across demographic groups, body locations, and lesion classes. Even when the metadata is not directly used as model input, it can still provide valuable information about class imbalance, missing values, and potential dataset biases.

From a quantitative point of view, the dataset is relatively large for a medical image classification task. It contains **25,331 training images** and **8,238 test images**, making it substantially larger than many earlier dermatology benchmarks. This scale is important because deep learning models, especially CNNs, typically benefit from larger datasets that expose them to greater visual variability. At the same time, dataset size alone does not make the task easy. The classes are not uniformly distributed, and some lesion categories

are represented by many more examples than others. Therefore, although the dataset is large enough to support supervised training, it still poses a challenging imbalanced multi-class classification problem.

The classification task in this project is a **9-class skin-lesion classification problem**. This means that the model must assign each image to one out of nine possible diagnostic categories. Compared with binary classification, where the goal is simply to separate benign from malignant lesions, a 9-class setting is significantly more demanding. The model must learn finer distinctions between lesion types, some of which may share similar visual characteristics. This makes the task more clinically realistic, because real dermatological diagnosis involves differentiating among several lesion categories rather than only detecting malignancy.

The nine classes cover a diverse range of lesion types, including both benign and malignant categories. As a consequence, the classification boundaries are often complex. Some classes are visually distinctive, while others overlap considerably in appearance. Furthermore, certain categories have many more images than others, which can bias the model toward frequent classes if no corrective strategies are adopted. For this reason, evaluating the model only with overall accuracy would not be sufficient. Class-wise precision, recall, and F1-score are also necessary in order to assess whether the network performs consistently across all lesion categories and not only on the dominant ones.

The metadata file provides a useful complementary view of the dataset structure. Typical metadata fields include:

- a unique image or lesion identifier;
- the diagnostic label associated with the image;
- patient demographic variables such as age and sex;
- anatomical site information indicating where the lesion is located on the body;

These metadata fields are valuable for several reasons. First, they allow a clearer understanding of how representative the dataset is across patient groups and lesion locations. Second, they can reveal missing or unevenly distributed information, which is common in medical datasets. Third, they help support visual and statistical exploration before model training. For example, age distributions may differ across lesion classes, some anatomical sites may be overrepresented, and some metadata categories may contain missing values. All of these aspects are relevant when interpreting model results, because they may influence both training behavior and generalization.

Overall, the ISIC 2019 dataset provides a strong foundation for this project because it combines a substantial number of dermoscopic images with useful tabular metadata in a realistic multi-class setting. Its scale makes deep learning feasible, while its imbalance, visual complexity, and class overlap make the problem genuinely challenging. These characteristics make it an appropriate benchmark for evaluating an efficiency-aware CNN, since the model must not only learn discriminative visual features, but also remain robust under realistic medical-data conditions.

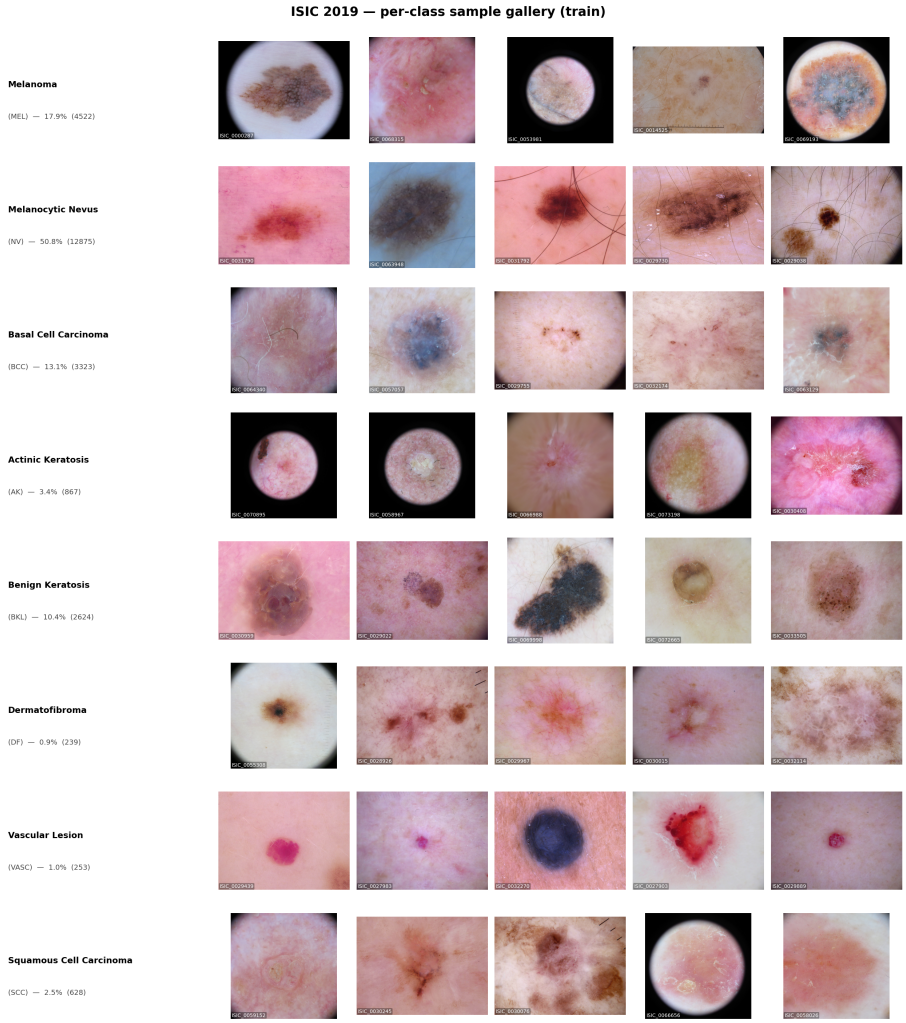


Figure 2.1: Representative dermoscopic images from the ISIC 2019 dataset across different lesion classes. The figure highlights visual variability, inter-class similarity, and common artifacts present in the data.

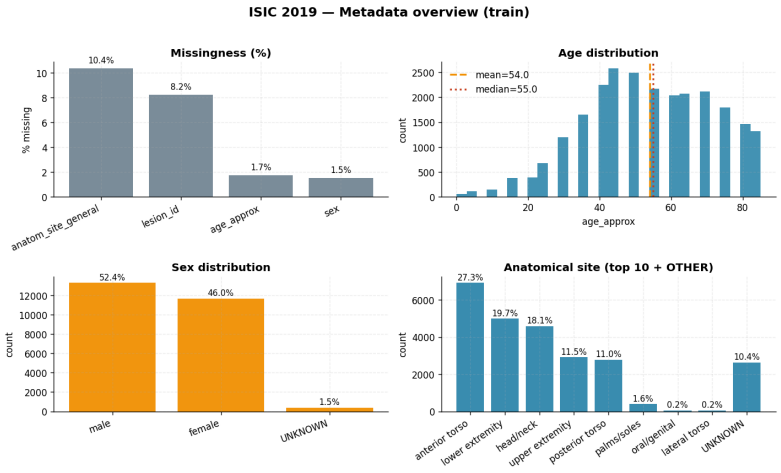


Figure 2.2: Overview of the dataset metadata, including example descriptive charts such as class distribution, patient sex, age distribution, and anatomical-site frequencies. These plots help illustrate class imbalance and the overall structure of the dataset.

Chapter 3

Major Issues in the Dataset

3.1 UNK Class and Class Imbalance

One of the first major issues identified in the dataset concerns the presence of the `UNK` class and the strong imbalance across the remaining diagnostic categories. Both aspects are important because they directly affect the reliability of training, the interpretation of performance, and the fairness of evaluation across classes.

The `UNK` class represents samples with uncertain, ambiguous, or insufficiently specific diagnostic information. From a machine learning perspective, including such a category in a supervised classification task creates an important problem: the label does not correspond to a coherent visual concept in the same way as the other diagnostic classes. In other words, while classes such as melanoma or nevus refer to lesion types with some degree of clinical and visual consistency, the `UNK` category acts more as a residual group than as a true semantic class. As a consequence, keeping it in the dataset may introduce label noise and make the task less well defined.

For this reason, the `UNK` class was removed from the main classification pipeline. This choice improves the clarity of the learning objective, since the model is trained only on classes that correspond to more meaningful diagnostic categories. Removing `UNK` also makes the evaluation more interpretable: a correct prediction can then be understood as assigning the lesion to a clinically defined class rather than to a vague fallback category. Although this reduces the total number of usable samples, it results in a cleaner and more consistent multi-class problem.

After removing `UNK`, the dataset still remains strongly imbalanced. Some lesion categories contain a large number of examples, while others are represented by much smaller subsets. This is a common issue in medical datasets, where frequent benign lesions tend to dominate and rarer or clinically critical classes appear much less often. Such imbalance can bias the optimization process because the model is exposed much more frequently to majority classes during training. If no corrective strategy is considered, the network may achieve apparently good overall accuracy simply by performing well on dominant categories while underperforming on minority ones.

Figure 3.1 shows the class distribution after preprocessing. The figure highlights the asymmetry of the dataset and makes clear that the classification problem cannot be treated as if all classes contributed equally. In practical terms, this means that the model must be evaluated with caution: aggregate accuracy alone would provide an incomplete and potentially misleading picture of performance.

Because of this imbalance, balanced evaluation metrics are especially important. In

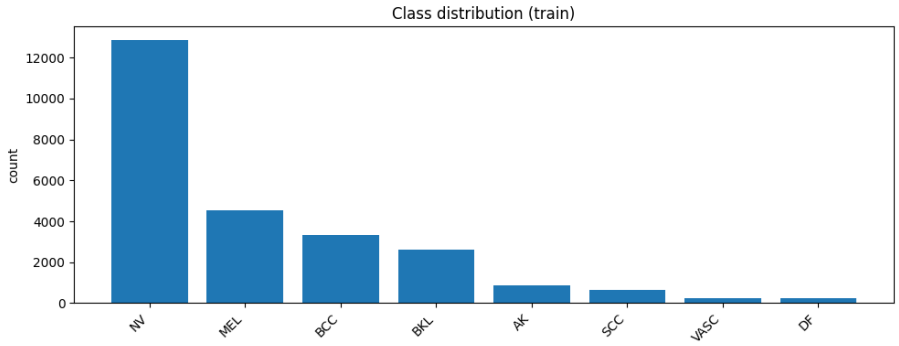


Figure 3.1: Class distribution of the dataset after preprocessing. The figure highlights the strong imbalance across lesion categories.

a highly skewed dataset, overall accuracy is dominated by the largest classes and may hide poor recognition of underrepresented categories. For this reason, metrics such as macro precision, macro recall, and especially macro F1-score are more informative. Macro F1 gives equal importance to each class by computing the F1-score independently per class and then averaging across classes. This prevents large classes from overwhelming the evaluation and provides a more realistic measure of whether the classifier performs consistently across the full diagnostic spectrum.

This consideration is particularly relevant in medical image classification. A model that performs very well on common benign lesions but fails on rarer or more clinically significant categories would not be satisfactory, even if its global accuracy appeared high. Therefore, throughout this project, balanced metrics are used alongside accuracy in order to better capture the true quality of the classifier. In summary, the removal of `UNK` helps define a cleaner prediction task, while the strong class imbalance motivates the use of evaluation criteria that reflect performance across all classes rather than only the majority ones.

3.2 Image Quality Variability

A second major issue in the dataset is the high variability in image quality. Unlike controlled benchmark datasets where images are often standardized, dermoscopic images in ISIC 2019 are collected under realistic clinical conditions and therefore exhibit substantial heterogeneity. This variability is important because it affects both the difficulty of the classification task and the robustness required from the model.

Several forms of image-quality variation can be observed. First, lighting is not uniform across samples. Some images are well illuminated and show clear lesion boundaries, while others are darker, overexposed, or affected by shadows and uneven brightness. These differences may alter perceived color distribution and local contrast, both of which are clinically relevant visual cues. Second, framing is inconsistent: lesions may appear centered in some images but off-center in others, and the relative proportion of lesion area to background skin may vary considerably. Third, image resolution and sharpness are not constant. Certain images are highly detailed, whereas others appear blurrier or lower in quality, which can make subtle lesion structures harder to detect.

In addition to these sources of variability, the dataset also contains outlier cases and visual artifacts. Examples include hair crossing the lesion, ruler marks, color calibration

patches, dark borders, air bubbles, and unusual background patterns. While these artifacts are realistic and reflect real clinical data acquisition, they introduce nuisance factors that are unrelated to the diagnostic category. If the model is not properly regularized, it may partially rely on these spurious cues instead of focusing on the lesion itself.

Figure 3.2 illustrates the diversity in image quality and presentation across the dataset. The examples show how differences in brightness, framing, sharpness, and artifact presence can make the classification task more challenging. This heterogeneity is one of the reasons why simple memorization is not sufficient: the network must learn visual features that remain useful across many acquisition conditions.

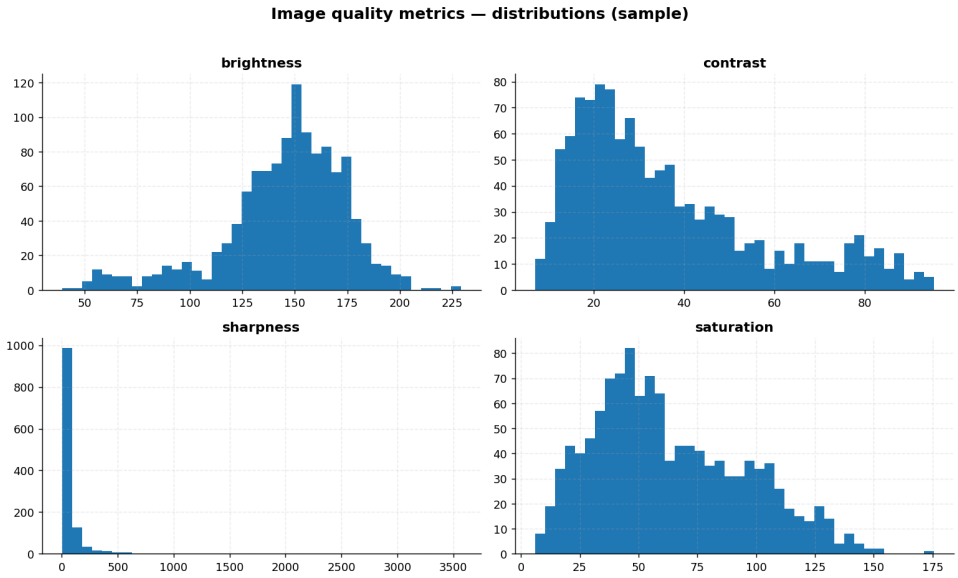


Figure 3.2: Examples illustrating image-quality variability in the dataset, including differences in illumination, framing, sharpness, and artifacts.

To reduce the negative impact of this variability, a normalization strategy is required. In this project, normalization is applied as part of the preprocessing pipeline in order to make the image input more consistent across samples. First, all images are resized to a common spatial resolution so that they can be processed in batches and fed into the CNN architecture under a fixed input format. This step reduces variation caused purely by image size. Second, pixel values are normalized so that the input distribution becomes more stable during optimization. This helps training converge more reliably and prevents differences in raw intensity scale from becoming an unnecessary source of noise.

Beyond standard normalization, data augmentation also plays an important role in dealing with image-quality variability. Random transformations such as horizontal flipping, small rotations, and mild color or geometric perturbations help expose the model to plausible variations that may occur in the real dataset. Although augmentation does not eliminate the underlying heterogeneity, it improves robustness by discouraging the network from depending too strongly on fragile low-level details.

Overall, image-quality variability is a central challenge of the ISIC 2019 dataset. It makes the task more realistic, but also more difficult, because the classifier must distinguish medically relevant visual patterns from irrelevant variation introduced by acquisition conditions. A careful normalization and augmentation pipeline is therefore essential for building a model that generalizes beyond the cleanest and most standardized cases.

3.3 Lesion ID and Data Leakage Risk

A third critical issue in the dataset concerns lesion identity and the associated risk of data leakage. In medical imaging datasets, different images do not always correspond to different underlying lesions. In ISIC 2019, a single lesion may appear in multiple images, for example because it was captured from slightly different angles, at different zoom levels, or under different acquisition conditions. This creates a potential problem when dividing the dataset into training and validation subsets.

If images of the same lesion are split across both training and validation sets, the model may effectively be evaluated on highly similar samples that it has already seen during training. Even if the exact pixels are different, repeated views of the same lesion can share many lesion-specific characteristics. This leads to data leakage, because the validation performance would no longer reflect true generalization to unseen lesions. Instead, it would partially measure the model's ability to recognize repeated instances of lesions already observed during training.

For this reason, lesion multiplicity must be analyzed explicitly. The presence of repeated lesion identifiers means that the natural unit of independence is not always the image, but rather the lesion itself. A dataset split that ignores this structure may overestimate model quality. Therefore, before training, it is important to inspect how many images correspond to unique lesion IDs, how many lesions appear more than once, and how these repetitions are distributed.

Figures 3.3 and 3.4 summarize this issue from two perspectives. The first figure provides an overview of lesion-ID multiplicity, showing how often lesions are represented by more than one image. The second figure highlights the presence of missing lesion identifiers, which is also relevant because missing IDs complicate strict group-based splitting and require careful handling during preprocessing.

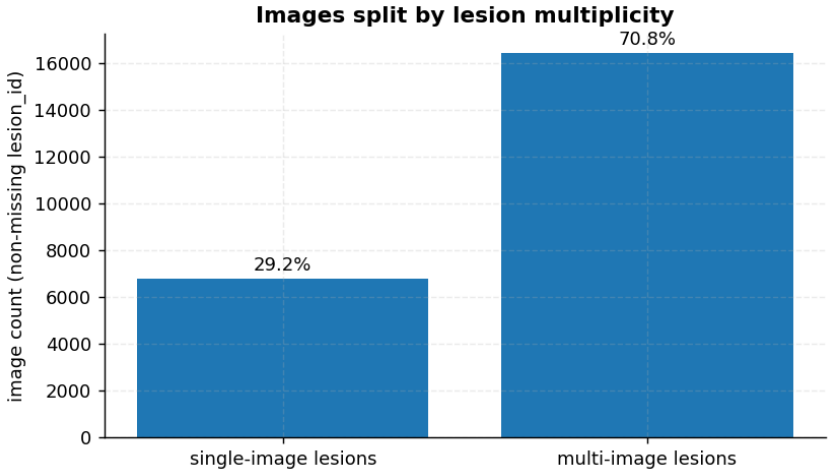


Figure 3.3: Overview of lesion-ID multiplicity, illustrating how many images correspond to repeated lesion instances.

To mitigate data leakage, this project adopts a lesion-aware splitting strategy based on `StratifiedGroupKFold`. This method is well suited to the problem because it addresses two requirements simultaneously. On the one hand, the grouping mechanism ensures that all images associated with the same lesion ID remain in the same fold, preventing the same lesion from appearing in both training and validation subsets. On the other hand, the

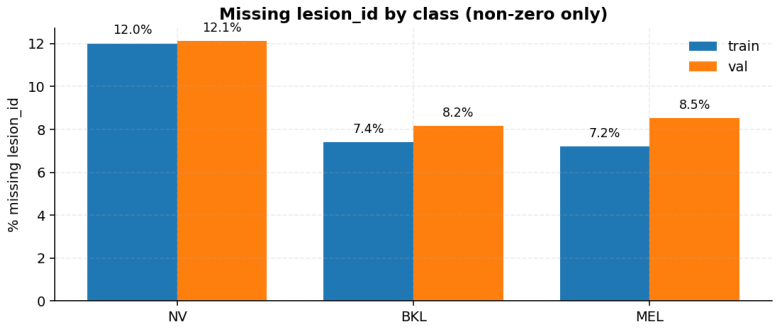


Figure 3.4: Summary of missing lesion identifiers in the dataset. Missing IDs require careful handling when constructing leakage-safe splits.

stratification mechanism attempts to preserve the class distribution across folds, which is particularly important given the strong imbalance discussed earlier.

This approach is preferable to a standard random split at image level. A random split may appear acceptable from a purely numerical point of view, but it does not respect the grouped structure of the data and can therefore produce overly optimistic validation results. By contrast, a lesion-aware split provides a stricter and more realistic estimate of generalization, since the model is validated on lesions that are truly unseen during training.

The use of `StratifiedGroupKFold` also aligns with the broader goal of this project, which is to obtain reliable rather than artificially inflated performance estimates. In medical classification, avoiding leakage is especially important because even small methodological errors can lead to misleading conclusions about model effectiveness. A model that seems strong under a leaky split may fail to generalize in practice when presented with entirely new lesions from real patients.

In summary, lesion-ID multiplicity introduces a significant leakage risk if dataset splitting is performed at image level without considering lesion grouping. A lesion-aware validation protocol is therefore necessary to ensure that the reported metrics reflect genuine generalization. By using `StratifiedGroupKFold`, the project preserves both lesion-level separation and approximate class balance, making the evaluation procedure more robust and trustworthy.

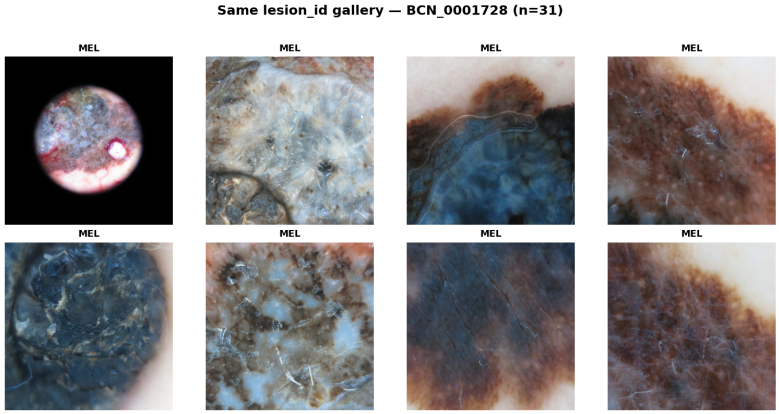


Figure 3.5: Examples of different images in the dataset with the same lesion id corresponding to the same skin lesion .

Chapter 4

Baseline Model Comparison

To determine the most suitable backbone for our efficiency-aware CNN, we evaluated three widely used lightweight architectures under identical raw training conditions: ResNet-18, MobileNetV2, and EfficientNet-B0. All models were trained for 15 epochs on 224×224 RGB images (ImageNet normalization), using StratifiedGroupKFold splitting (to prevent lesion-ID leakage), CrossEntropyLoss, Adam optimizer (learning rate 10^{-4}), and no data augmentation. This controlled setup enabled a fair comparison of their inherent generalization capability on the highly imbalanced ISIC 2019 dataset (8 classes after UNK removal).

Performance is assessed via training/validation loss curves and validation accuracy/-macro F1-score curves. Macro F1 is particularly important because of the severe class imbalance (NV dominates at ∼51% of samples).

4.1 ResNet-18 Baseline

ResNet-18, introduced by He et al. [3], is a classic residual network that employs skip connections to mitigate the vanishing gradient problem and enable deeper architectures. It consists of 18 layers with residual blocks using 3×3 convolutions and batch normalization. Its main strengths are simplicity, strong feature extraction through residual learning, and ease of training. However, its drawbacks include a relatively large number of parameters (∼11.7 million) and limited efficiency in terms of FLOPs compared to more modern lightweight models, making it less ideal for resource-constrained medical imaging applications [3].

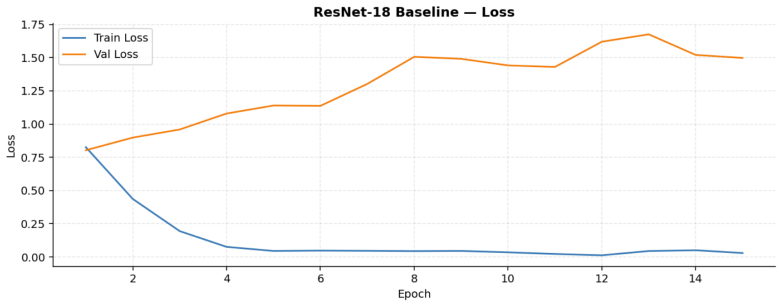


Figure 4.1: ResNet-18: Training and validation loss curves.

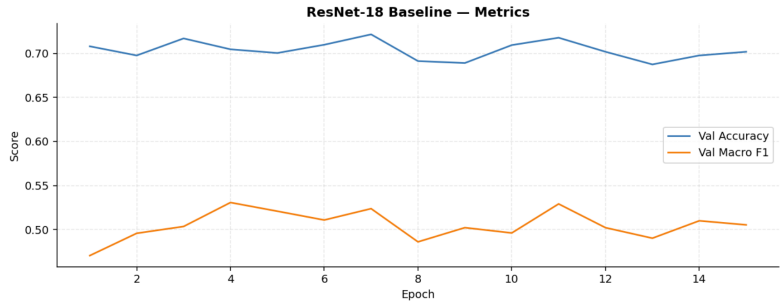


Figure 4.2: ResNet-18: Validation accuracy and macro F1-score.

In our raw baseline experiments, ResNet-18 exhibited rapid convergence of the training loss. However, as shown in Figure 4.1, the validation loss began to rise after epoch 6, indicating clear overfitting. The validation accuracy stabilized around 0.70, while the macro F1-score remained low at approximately 0.51 (Figure 4.2). This behaviour reflects ResNet-18's tendency to overfit on the dominant class (NV) while struggling with minority classes, resulting in limited generalization on the imbalanced ISIC dataset.

4.2 MobileNetV2 Baseline

MobileNetV2, proposed by Sandler et al. [5], is a lightweight architecture designed for mobile and embedded vision applications. It introduces inverted residual blocks with linear bottlenecks and depthwise separable convolutions, significantly reducing parameters (∼3.4 million) and computational cost while maintaining reasonable accuracy. Its strengths lie in high efficiency and low latency. However, its shallower design and reliance on depthwise convolutions can limit its representational power on complex medical images with subtle inter-class differences, often leading to weaker generalization compared to compound-scaled models [5].

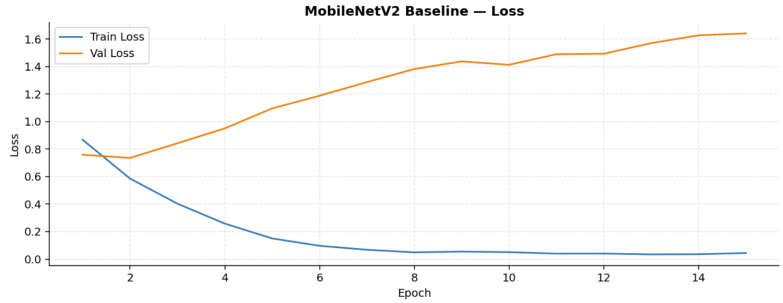


Figure 4.3: MobileNetV2: Training and validation loss curves.

Under the same raw training settings, MobileNetV2 achieved lower training loss than ResNet-18, yet the validation loss continued to increase throughout training (Figure 4.3). Validation accuracy reached approximately 0.73, but the macro F1-score stayed modest at around 0.55 (Figure 4.4). The widening train–validation gap highlights MobileNetV2's limited capacity to generalize on the ISIC dataset, confirming its tendency toward overfitting despite its efficiency advantages.

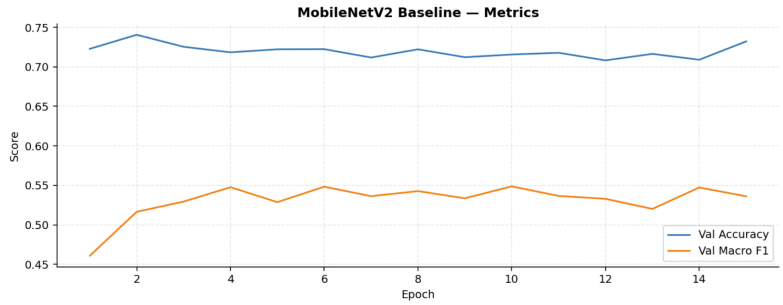


Figure 4.4: MobileNetV2: Validation accuracy and macro F1-score.

4.3 EfficientNet-B0 Baseline

EfficientNet-B0, introduced by Tan and Le [4], employs a compound scaling method that uniformly scales depth, width, and resolution using a simple yet effective coefficient. Built on mobile inverted bottleneck blocks with squeeze-and-excitation modules, it achieves state-of-the-art accuracy with significantly fewer parameters (∼5.3 million) and FLOPs than previous models. Its primary strengths are superior efficiency–accuracy trade-off and strong feature representation. Weaknesses are minimal for this task, though it still benefits from additional regularization techniques on highly imbalanced medical datasets [4].

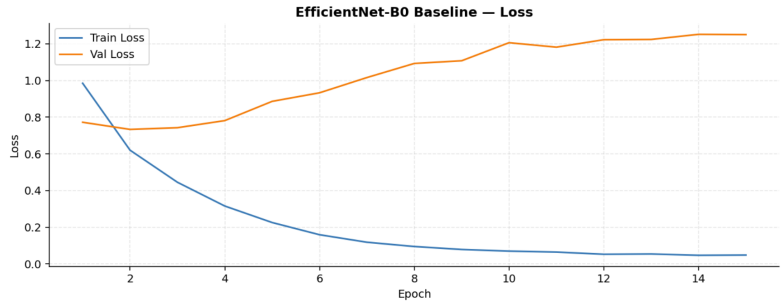


Figure 4.5: EfficientNet-B0: Training and validation loss curves.

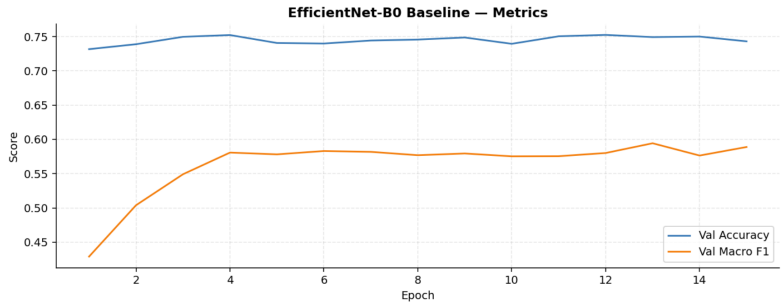


Figure 4.6: EfficientNet-B0: Validation accuracy and macro F1-score.

In our experiments, EfficientNet-B0 demonstrated the most stable training dynamics. Both training and validation losses decreased smoothly and remained closely aligned (Figure 4.5). It achieved the highest validation accuracy (≈0.75) and the best macro F1-score (≈0.58) among the three baselines (Figure 4.6). The flat and consistent validation

curves indicate excellent generalization, even on the imbalanced ISIC data, making it the strongest candidate for further efficiency-aware enhancements.

4.4 Comparison Summary

Table 4.1: Baseline comparison under identical raw training settings (15 epochs, no augmentation).

Model	**Final Val Loss**	**Val Accuracy**	**Macro F1**	**Key Observation**
ResNet-18	∼1.50	∼0.70	∼0.51	Clear overfitting after epoch 6
MobileNetV2	∼1.62	∼0.73	∼0.55	Largest train–validation gap
EfficientNet-B0	**∼1.20**	**∼0.75**	**∼0.58**	**Best and most stable performance**

EfficientNet-B0 clearly outperformed ResNet-18 and MobileNetV2, delivering higher validation accuracy, higher macro F1, and the most stable validation curves. ResNet-18 and MobileNetV2 exhibited weaker generalization and more pronounced overfitting on the minority classes. Consequently, EfficientNet-B0 was selected as the baseline model for all subsequent improvements (light data augmentation, CutMix, cosine annealing scheduler, and TTA).

All three baselines were implemented using the same training pipeline, as detailed in the attached notebooks (`ISIC_Baseline_ResNet_18.ipynb`, `ISIC_Baseline_mobilenet.ipynb`, and `Isic_efficient_net_baseline.ipynb`).

Chapter 5

Improvement Strategy

Based on the experimental results obtained from the ISIC 2019 dataset, the baseline model provided a reasonable foundation for skin lesion classification. However, to ensure suitability for real-world clinical applications, further improvements were necessary to enhance both predictive performance and computational efficiency.

The dataset presented several inherent challenges. A portion of the images suffered from poor lighting conditions, motion blur, occlusion by hair, and other visual artifacts, which limited the model's ability to capture discriminative features. Moreover, the dataset exhibited significant class imbalance, with certain lesion categories containing thousands of samples while others were severely underrepresented. This imbalance adversely affected the model's generalization performance, particularly for minority classes.

The primary objective of the proposed improvement strategies was to increase classification accuracy while maintaining or reducing computational complexity. By addressing data-related limitations and optimizing the training process, the model was expected to generalize more effectively to previously unseen images. Four key improvement strategies were implemented: light data augmentation, CutMix regularization, cosine annealing learning rate scheduling, and test-time augmentation.

5.1 Light Data Augmentation

Data augmentation was employed to artificially increase the diversity of the training dataset. Instead of repeatedly presenting identical images to the model, random horizontal flipping and small-angle rotations were applied to generate slightly modified versions of the original images.

This approach encourages the model to learn invariant and robust feature representations. For instance, a skin lesion remains diagnostically identical regardless of orientation. Augmentation therefore enables the model to recognize pathological features under varying spatial configurations and lighting conditions. As a result, it improves generalization and reduces overfitting, particularly in scenarios where the original dataset contains inconsistencies in image acquisition.

5.2 CutMix

CutMix is a data augmentation and regularization technique in which a patch from one image is randomly cropped and pasted onto another image. The corresponding labels are

proportionally mixed according to the area of the inserted patch. For example, if two images contribute equally to the final composite image, the associated label is represented as a weighted combination of both classes.

This technique encourages the model to consider the entire image rather than focusing on a limited local region. In medical image analysis, such behavior is especially beneficial, as pathological features may appear in various locations and may be partially obscured. CutMix promotes spatial awareness and discourages the model from memorizing isolated patterns or artifacts. Consequently, it enhances generalization performance and reduces the risk of overfitting.

Listing 5.1: CutMix Implementation

```
CUTMIX_PROB = 0.5
CUTMIX_ALPHA = 1.0

def rand_bbox(size, lam):
    W = size[2]
    H = size[3]
    cut_rat = np.sqrt(1.0 - lam)
    cut_w = int(W * cut_rat)
    cut_h = int(H * cut_rat)

    cx = np.random.randint(W)
    cy = np.random.randint(H)

    x1 = np.clip(cx - cut_w // 2, 0, W)
    y1 = np.clip(cy - cut_h // 2, 0, H)
    x2 = np.clip(cx + cut_w // 2, 0, W)
    y2 = np.clip(cy + cut_h // 2, 0, H)

    return x1, y1, x2, y2

def apply_cutmix(images, labels, alpha=1.0):
    lam = np.random.beta(alpha, alpha)
    rand_index = torch.randperm(images.size(0), device=images.device)

    target_a = labels
    target_b = labels[rand_index]

    x1, y1, x2, y2 = rand_bbox(images.size(), lam)
    images[:, :, x1:x2, y1:y2] = images[rand_index, :, x1:x2, y1:y2]

    lam = 1 - ((x2 - x1) * (y2 - y1) / (images.size(-1) * images.size
        (-2)))
    return images, target_a, target_b, lam
```

5.3 Cosine Annealing Scheduler

A cosine annealing learning rate scheduler was adopted to optimize the training dynamics. Rather than maintaining a constant learning rate throughout training, the scheduler gradually reduces the learning rate following a cosine decay curve.

At the early stages of training, a relatively high learning rate enables the model to rapidly capture global and coarse-grained patterns, such as overall color distributions and structural characteristics of different lesion types. As training progresses, the learning

rate decreases smoothly, allowing the model to fine-tune subtle and discriminative features that differentiate visually similar classes.

This gradual reduction prevents abrupt parameter updates during later training stages, thereby improving training stability and convergence behavior. Ultimately, the cosine annealing strategy contributes to improved model robustness and more precise classification performance.

Listing 5.2: Cosine Annealing Scheduler

```
optimizer = optim.Adam(model.parameters(), lr=1e-4)

scheduler = torch.optim.lr_scheduler.CosineAnnealingLR(
    optimizer,
    T_max=15,
    eta_min=1e-6
)
```

5.4 Test-Time Augmentation (TTA)

Test-Time Augmentation was applied during the inference phase to enhance prediction reliability. For each test image, several augmented versions (e.g., horizontally flipped or slightly rotated variants) were generated. The model produced predictions for each version, and the final output was obtained by averaging these predictions.

This strategy mitigates the impact of incidental visual distortions, such as blur, uneven illumination, or partial occlusion. By aggregating predictions from multiple perspectives of the same image, TTA reduces sensitivity to minor variations and improves robustness. In medical imaging contexts, where diagnostic accuracy is critical, this ensemble-like approach helps ensure that the final prediction reflects comprehensive feature evaluation rather than reliance on a single potentially imperfect representation.

Listing 5.3: TTA transform and inference

```
def tta_transforms(img):
    variants = []

    # original
    variants.append(preprocess_img(img))

    # horizontal flip
    variants.append(preprocess_img(TF.hflip(img)))

    # slight rotations
    variants.append(preprocess_img(TF.rotate(img, 10)))
    variants.append(preprocess_img(TF.rotate(img, -10)))

    return variants

model.eval()
all_preds_tta = []
all_labels = []

with torch.no_grad():
    for pil_images, labels in tta_loader:
        batch_probs = []

        for img in pil_images:
            variants = tta_transforms(img)
            variants = torch.stack(variants).to(device)   # [n_aug, C,
               H, W]

            outputs = model(variants)
            probs = torch.softmax(outputs, dim=1).mean(dim=0)  #
               average over augmentations
            batch_probs.append(probs)

        batch_probs = torch.stack(batch_probs)  # [B, num_classes]
        preds = torch.argmax(batch_probs, dim=1)

        all_preds_tta.extend(preds.cpu().numpy())
        all_labels.extend(labels.numpy())
```

Chapter 6

Final Results

In this section, we present the final results of our model's performance on the ISIC 2019 dataset, showcasing the effectiveness of our approach in classifying skin lesions. We used EfficientNet-B0 as the baseline model, enhancing it with test-time augmentation (TTA) to improve both predictive accuracy and generalization capabilities.

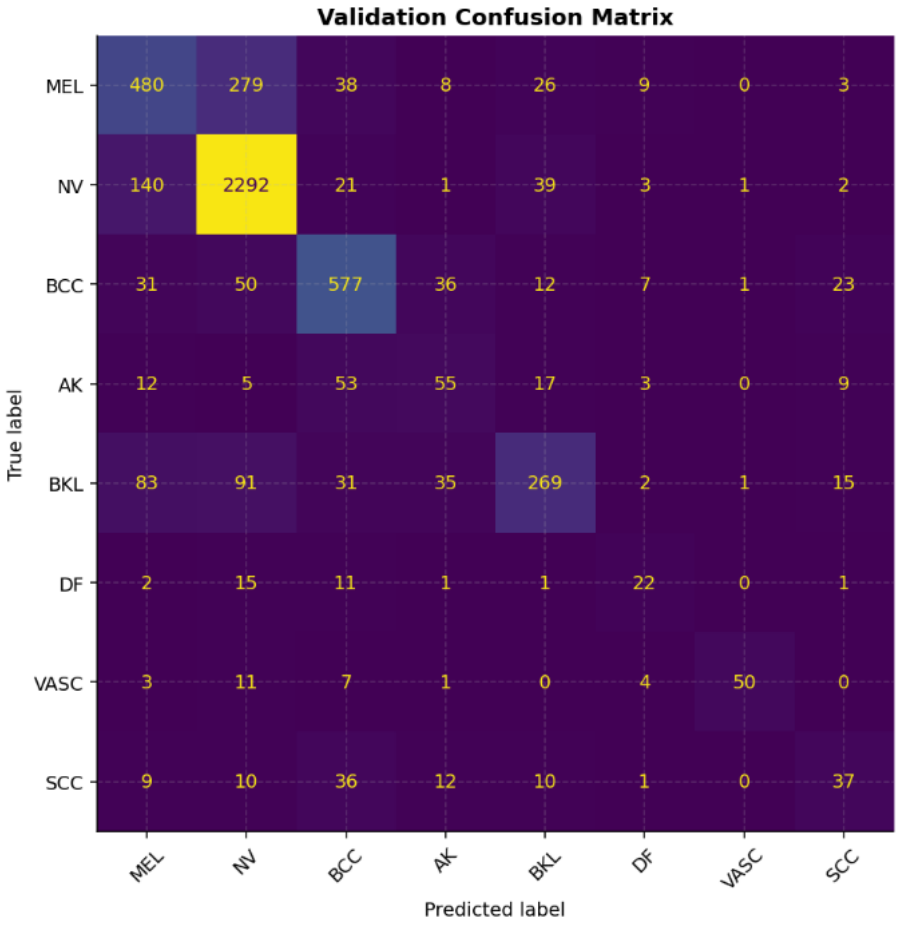


Figure 6.1: Validation Confusion Matrix (TTA).

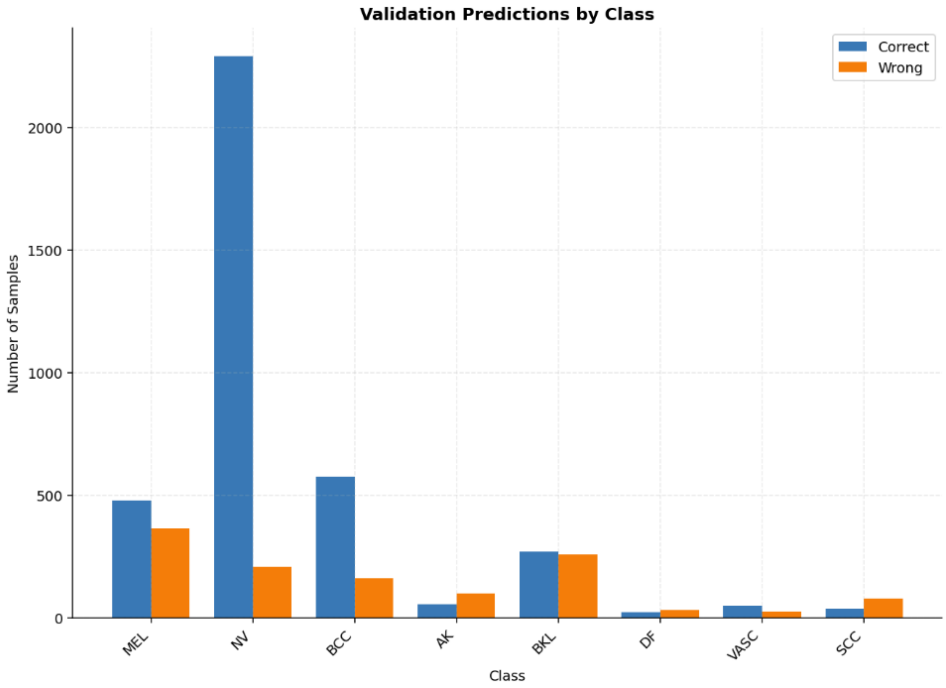


Figure 6.2: Validation Prediction by Class.

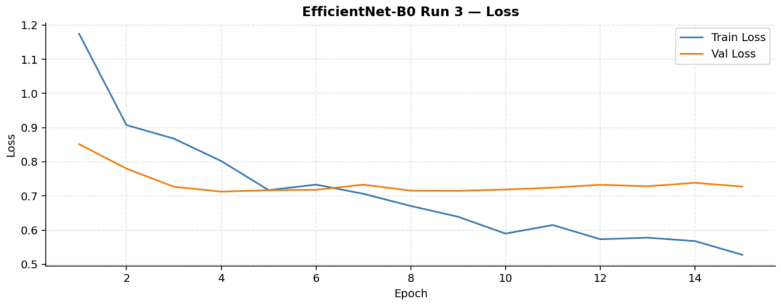


Figure 6.3: EfficientNet-B0 Improved: Training and validation loss curves.

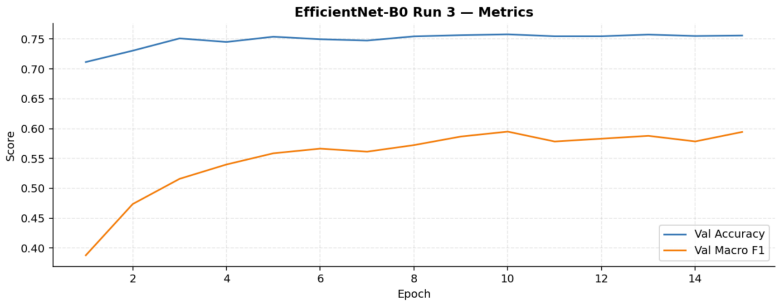


Figure 6.4: EfficientNet-B0 Improved: Validation accuracy and macro F1-score.

Listing 6.1: TTA Result

```
TTA Validation Accuracy: 0.7660
TTA Validation Macro F1: 0.6083

Classification Report (TTA):
              precision    recall  f1-score   support

         MEL     0.6633    0.5421    0.5966       843
          NV     0.8287    0.9348    0.8785      2499
         BCC     0.7594    0.7965    0.7775       737
          AK     0.3673    0.3506    0.3588       154
         BKL     0.7383    0.5408    0.6243       527
          DF     0.4400    0.4151    0.4272        53
        VASC     1.0000    0.6579    0.7937        76
         SCC     0.4667    0.3652    0.4098       115

    accuracy                         0.7660      5004
   macro avg     0.6580    0.5754    0.6083      5004
weighted avg     0.7571    0.7660    0.7565      5004
```

6.1 Validation Confusion Matrix (TTA)

The confusion matrix (Figure 6.1) provides a detailed view of our model's performance across the nine lesion classes. The matrix reveals that the model performed exceptionally well on common lesion types, such as NV (Nevus), BCC (Basal Cell Carcinoma), and MEL (Melanoma), with a high number of correct predictions. However, there was a noticeable drop in performance for less common categories such as AK (Actinic Keratosis), DF (Dermatofibroma), and SCC (Squamous Cell Carcinoma). This imbalance is expected given the scarcity of samples in these minority classes. Despite this, the model still shows a strong ability to predict common classes, making it useful in clinical settings where the majority of cases are benign.

6.2 Validation Prediction by Class

Figure 6.2 illustrates the validation predictions by class, where the correct predictions are shown in blue, and the incorrect ones are represented in orange. The model performed well on NV, BCC, and MEL, as evidenced by the high number of correct predictions. For minority classes like AK and DF, however, the number of incorrect predictions was relatively higher, indicating the challenges associated with underrepresented categories.

6.3 Loss and F1-Score Curves

Figures 6.3 and 6.4 show the training and validation loss curves and the validation accuracy and macro F1-score respectively. The results for EfficientNet-B0 with TTA demonstrate stable loss reduction across both training and validation sets, indicating good generalization. The validation accuracy achieved was 76.60, with the macro F1-score reaching 60.83. The improvement in F1-score is crucial, as it balances the performance across all classes, ensuring that the model is not overly biased towards the dominant classes.

6.4 Classification Report and Class-wise Performance

In Listing 6.1, we provide a detailed classification report that includes precision, recall, and F1-score metrics for each class. The model achieved impressive performance on NV, with a precision of 82.87, recall of 93.48, and F1-score of 87.85. However, for AK, DF, and SCC, the F1-scores were significantly lower, with AK showing an F1-score of 35.88. This highlights the ongoing challenge of class imbalance, where classes with fewer examples are more difficult for the model to learn.

6.5 Future Work

The results demonstrate that our model, based on EfficientNet-B0 with TTA, forms a solid baseline for skin lesion classification, with a strong performance on the majority classes. Despite this, the model struggles with minority classes, which reflects the underlying challenges of working with imbalanced datasets in medical imaging.

In clinical applications, this model could provide a useful diagnostic tool, especially for more prevalent skin lesions, but further work is needed to improve the performance on rarer lesion types. Future research could focus on integrating advanced class imbalance mitigation techniques, attention-based mechanisms, or incorporating multimodal learning that combines clinical metadata with imaging data. By addressing these areas, we aim to enhance the model's sensitivity to minority classes, improving its robustness and overall clinical applicability.

This work contributes to the growing field of AI in dermatology, underlining its potential as a decision-support tool in medical diagnostics.

Chapter 7

Conclusion

This study presents a comprehensive approach to skin lesion classification using deep learning techniques, specifically convolutional neural networks (CNNs). The primary goal of this work was to design a model that balances diagnostic performance with computational efficiency, making it suitable for deployment in real-world clinical environments, particularly where resources may be limited. By leveraging the ISIC 2019 dataset, which includes a wide range of skin lesion images, we addressed several key challenges and proposed strategies for improving classification accuracy while maintaining efficiency.

7.1 Key Findings

Skin lesion classification is a clinically significant task, particularly for the early detection of skin cancer, which can drastically improve patient outcomes. The ISIC 2019 dataset provided a diverse and complex set of images, representative of the challenges faced in dermatology. The visual similarities among different types of lesions and the high variability in image quality made the task particularly difficult. In this context, deep learning models, particularly CNNs, offer a promising solution by automatically learning discriminative features from raw image data.

Throughout the study, we focused on the development of an efficiency-aware CNN, emphasizing the practical aspect of model deployment. Our baseline model, EfficientNet-B0, was selected due to its ability to provide a good trade-off between accuracy and computational efficiency. This model was enhanced with several key techniques to further improve its robustness and generalization performance.

7.2 Challenges Addressed

Several inherent challenges were addressed during this study, including:

Class Imbalance: The ISIC 2019 dataset is highly imbalanced, with some lesion categories significantly outnumbering others. This imbalance posed a risk of bias in the model's predictions, especially for minority classes. To address this, we used balanced evaluation metrics such as macro F1-score, which provides a more comprehensive view of the model's performance across all classes, rather than focusing solely on accuracy.

Image Quality Variability: The dataset consisted of dermoscopic images captured under varying conditions, including differences in lighting, sharpness, framing, and the presence of artifacts such as hair or rulers. This variability can significantly impact the

model's ability to learn meaningful features. To mitigate these issues, we implemented a preprocessing pipeline that included normalization and data augmentation techniques to expose the model to a wider range of image variations.

Lesion Identity and Data Leakage: A critical issue in medical imaging datasets is the risk of data leakage due to repeated images of the same lesion. This can lead to overestimated validation performance if images of the same lesion are included in both the training and validation sets. To prevent this, we employed the StratifiedGroupKFold strategy, which ensures that images from the same lesion are grouped together in the training or validation sets, thereby preserving the independence of the test data.

The confusion matrix and classification report further emphasized the model's strengths and weaknesses. While the model performed well on the dominant classes, it struggled with the minority classes, as evidenced by the lower F1-scores for AK and DF. This indicates that the model is more biased towards common lesions, and additional efforts will be required to improve its performance on rarer classes.]Model Performance and Results Our final model, which combined EfficientNet-B0 with test-time augmentation (TTA), achieved a validation accuracy of 76.60 and a macro F1-score of 60.83. The model showed strong performance on common lesion types like NV (Nevus), BCC (Basal Cell Carcinoma), and MEL (Melanoma), with high precision and recall. However, the performance on minority classes such as AK (Actinic Keratosis), DF (Dermatofibroma), and SCC (Squamous Cell Carcinoma) was less impressive, with lower recall and F1-scores for these categories. These results highlight the ongoing challenge of dealing with class imbalance in medical image classification tasks.

The confusion matrix and classification report further emphasized the model's strengths and weaknesses. While the model performed well on the dominant classes, it struggled with the minority classes, as evidenced by the lower F1-scores for AK and DF. This indicates that the model is more biased towards common lesions, and additional efforts will be required to improve its performance on rarer classes.

7.3 Future Directions

While the model demonstrated a solid foundation for skin lesion classification, there are several potential areas for improvement. First, additional strategies for mitigating class imbalance, such as synthetic data generation or advanced re-sampling techniques, could help improve the model's sensitivity to minority classes. Techniques like focal loss, which places more emphasis on hard-to-classify examples, could also be explored to address the performance gaps in underrepresented categories.

Second, model interpretability remains an important research direction. Understanding how the model makes decisions, particularly in medical applications, is crucial for clinicians who may rely on the system for diagnostic support. Future research could incorporate attention mechanisms or use methods such as Grad-CAM to visualize the regions of the image that the model focuses on when making predictions.

Third, the integration of clinical metadata (such as patient age, sex, and lesion location) with image data could potentially improve the model's performance. Multimodal learning, where both clinical and imaging data are combined, could provide additional context and enhance the model's robustness, especially in real-world clinical settings where patient history plays a significant role in diagnosis.

7.4 Contribution to Dermatological AI Research

This study contributes to the growing body of research on artificial intelligence in dermatology, underscoring the potential of AI-driven tools to assist in the detection of skin cancer and other dermatological conditions. By focusing on efficiency, we demonstrated that a lightweight model could achieve competitive performance on a challenging medical imaging task. This has significant implications for real-world applications, where model deployment must be feasible in resource-constrained environments.

As AI continues to advance, the integration of automated classification systems into clinical workflows has the potential to support dermatologists by providing consistent, rapid, and scalable diagnostic assistance. In conclusion, this work serves as a valuable step toward the development of efficient AI systems for medical image analysis, with the potential to significantly improve patient outcomes in dermatology.

Bibliography

[1] ISIC 2019 Challenge Dataset.

[2] Mingxing Tan and Quoc V. Le, "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks," 2019.

[3] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 770–778).

[4] Tan, M., & Le, Q. V. (2019). EfficientNet: Rethinking model scaling for convolutional neural networks. In *International Conference on Machine Learning* (pp. 6105–6114).

[5] Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., & Chen, L.-C. (2018). MobileNetV2: Inverted residuals and linear bottlenecks. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 4510–4520).

Appendix A

Contribution Report

Hung Siu Wang (57148331) – Team Leader

Responsible for designing the overall project structure and coordinating team progress. Conducted research on the available ISIC dataset and selected three baseline models for comparative analysis. Implemented the baseline ResNet-18 model and completed Chapter 4 (Baseline Model Comparison) of the report. Also responsible for the final refinement and overall proofreading of the full report.

Matteo Zabban (40172615)

Implemented the data preprocessing pipeline, including exploratory data analysis (EDA) and data cleaning procedures. Developed the full training framework for the models, incorporating data augmentation, CutMix regularization, cosine annealing scheduler, and Test-Time Augmentation (TTA). Responsible for writing Chapters 1–3 and the Abstract of the report and the Demo Video.

Chan Tsz Hin (58539653)

Led the preparation and design of the presentation slides. Responsible for Chapter 6 (Final Results) of the report.

Choi San Ho (57849436)

Implemented the baseline MobileNetV2 model. Responsible for writing Chapters 5 (Improvement Strategy) and 7 (Conclusion) of the report.